

POISSON $Y_i = \beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i}$

$$f(y_i; \mu_i) = \frac{e^{-\mu_i} \mu_i^{y_i}}{y_i!} \quad \mu_i = e^{\tilde{x}_i^T \beta}$$

• LIKELIHOOD

$$L(\beta) = \prod \frac{e^{-\mu_i} \mu_i^{y_i}}{y_i!}$$

• LOG LIKELIHOOD

$$\begin{aligned} \lambda(\beta) &= \log(L(\beta)) = \log\left(\prod \frac{e^{-\mu_i} \mu_i^{y_i}}{y_i!}\right) = \sum \log\left(\frac{e^{-\mu_i} \mu_i^{y_i}}{y_i!}\right) = \sum \log(e^{-\mu_i}) + \log(\mu_i)^{y_i} - \log(y_i!) \\ &= \sum -\mu_i + y_i \log(\mu_i) - \log(y_i!) = \sum -e^{\tilde{x}_i^T \beta} + y_i \tilde{x}_i^T \beta - \log(y_i!) \end{aligned}$$

• SCORE FUNCTION

$$\lambda(\beta) = \sum -e^{\tilde{x}_i^T \beta} + y_i \tilde{x}_i^T \beta - \log(y_i!) = -\sum e^{\beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i}} + \sum y_i (\beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i}) - \log(y_i!)$$

$$\lambda(\beta) = \begin{cases} \alpha = \frac{d}{d\beta_1} \lambda(\beta) = \frac{d}{d\beta_1} \left[-\sum e^{\beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i}} + \sum y_i (\beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i}) - \log(y_i!) \right] = -\sum e^{\beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i}} + \sum y_i \\ \beta = \frac{d}{d\beta_2} \lambda(\beta) = -\sum e^{\beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i}} x_{1i} + \sum y_i x_{1i} = \sum x_{1i} (y_i - e^{\beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i}}) \\ \gamma = \frac{d}{d\beta_3} \lambda(\beta) = -\sum e^{\beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i}} x_{2i} + \sum y_i x_{2i} = \sum x_{2i} (y_i - e^{\beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i}}) \end{cases}$$

GAUSSIAN

$$f(y_i; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y_i - \mu_i)^2} \quad \mu_i = \beta_1 + \beta_2 x_i$$

LIKELIHOOD

$$L(\theta) = \prod \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}[y_i - \mu_i]^2\right\} = \prod \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}\sum [y_i - \mu_i]^2\right\} \\ = (2\pi)^{-n/2} (\sigma^2)^{-n/2} \exp\left\{-\frac{1}{2\sigma^2}\sum (y_i - \mu_i)^2\right\}$$

LOG-LIKELIHOOD

$$l(\theta) = \log(L(\theta)) = \log\left\{(2\pi)^{-n/2} (\sigma^2)^{-n/2} \exp\left\{-\frac{1}{2\sigma^2}\sum (y_i - \mu_i)^2\right\}\right\} = \log(2\pi)^{-n/2} + \log(\sigma^2)^{-n/2} - \frac{1}{2\sigma^2}\sum (y_i - \mu_i)^2 \\ = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum (y_i - \mu_i)^2$$

SCORE FUNCTION

$$l_{\theta}(\theta) = \begin{cases} a = \frac{d}{d\beta_1} l(\theta) \\ b = \frac{d}{d\beta_2} l(\theta) \\ x = \frac{d}{d\sigma^2} l(\theta) \end{cases}$$

$$a = \frac{d}{d\beta_1} l(\theta) = \frac{d}{d\beta_1} \left[-\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum (y_i - \mu_i)^2 \right] = \frac{d}{d\beta_1} \left[-\frac{1}{2\sigma^2}\sum (y_i - \beta_1 - \beta_2 x_i)^2 \right] = -\frac{1}{\sigma^2}\sum (y_i - \beta_1 - \beta_2 x_i)(-1) = \frac{1}{\sigma^2}\sum y_i - \beta_1 - \beta_2 x_i$$

$$b = \frac{d}{d\beta_2} l(\theta) = \frac{d}{d\beta_2} \left[-\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum (y_i - \mu_i)^2 \right] = \frac{d}{d\beta_2} \left[-\frac{1}{2\sigma^2}\sum (y_i - \beta_1 - \beta_2 x_i)^2 \right] = -\frac{1}{\sigma^2}\sum (y_i - \beta_1 - \beta_2 x_i)(x_i) = -\frac{1}{\sigma^2}\sum (y_i - \beta_1 - \beta_2 x_i)(x_i)$$

$$x = \frac{d}{d\sigma^2} l(\theta) = \frac{d}{d\sigma^2} \left[-\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum (y_i - \mu_i)^2 \right] = -\frac{n}{2} \frac{1}{\sigma^2} + \frac{1}{2}\sum (y_i - \mu_i)^2 \frac{1}{(\sigma^2)^2}$$

$$\frac{d}{dx} \frac{1}{x} = \frac{d}{dx} x^{-1} = -1 x^{-2} = -\frac{1}{x^2}$$

- GAUSSIAN: $f(y_i; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}[y_i - \mu_i]^2\right\}$ $\mu_i = \beta_0 + \beta_1 x_i$

- BERNOULLI: $f(y_i; \theta) = \pi_i^{y_i} (1 - \pi_i)^{1-y_i}$ $\pi_i = \frac{e^{\tilde{x}_i^T \theta}}{1 + e^{\tilde{x}_i^T \theta}}$

- POISSON $f(y_i; \theta) = \frac{e^{-\lambda_i} \lambda_i^{y_i}}{y_i!}$ $\lambda_i = e^{\tilde{x}_i^T \theta}$

BERNOULLI

$$f(y_i; \theta) = \pi_i^{y_i} (1 - \pi_i)^{1-y_i} \quad \pi_i = \frac{e^{\tilde{x}_i^T \theta}}{1 + e^{\tilde{x}_i^T \theta}}$$

Likelihood

$$L(\theta) = \prod \pi_i^{y_i} (1 - \pi_i)^{1-y_i}$$

LOG-Likelihood

$$\begin{aligned} \ell(\theta) = \log(L(\theta)) &= \log\left(\prod \pi_i^{y_i} (1 - \pi_i)^{1-y_i}\right) = \sum \log(\pi_i)^{y_i} + (1 - y_i) \log(1 - \pi_i) = \sum y_i \log\left(\frac{e^{\tilde{x}_i^T \theta}}{1 + e^{\tilde{x}_i^T \theta}}\right) + \log\left(\frac{1}{1 + e^{\tilde{x}_i^T \theta}}\right) - y_i \log\left(\frac{1}{1 + e^{\tilde{x}_i^T \theta}}\right) \\ &= \sum y_i \log(e^{\tilde{x}_i^T \theta}) - y_i \log(1 + e^{\tilde{x}_i^T \theta}) - \log(1 + e^{\tilde{x}_i^T \theta}) + y_i \log(1 + e^{\tilde{x}_i^T \theta}) = \sum y_i \tilde{x}_i^T \theta - \log(1 + e^{\tilde{x}_i^T \theta}) \end{aligned}$$

SCORE FUNCTION

$$s(\theta) = \begin{cases} a = \frac{d}{d\theta_1} \ell(\theta) \\ b = \frac{d}{d\theta_2} \ell(\theta) \end{cases}$$

$$\begin{cases} a = \frac{d}{d\theta_1} \left[\sum y_i \tilde{x}_i^T \theta - \log(1 + e^{\tilde{x}_i^T \theta}) \right] = \frac{d}{d\theta_1} \sum y_i (\beta_1 + \beta_2 x_i) - \log(1 + e^{\beta_1 + \beta_2 x_i}) = \sum y_i - \frac{e^{\beta_1 + \beta_2 x_i}}{1 + e^{\beta_1 + \beta_2 x_i}} \cdot 1 \\ b = \frac{d}{d\theta_2} \left[\sum y_i \tilde{x}_i^T \theta - \log(1 + e^{\tilde{x}_i^T \theta}) \right] = \frac{d}{d\theta_2} \sum y_i x_i - \frac{e^{\beta_1 + \beta_2 x_i}}{1 + e^{\beta_1 + \beta_2 x_i}} \cdot x_i \end{cases}$$

DISTRIBUTION OF ESTIMATORS

$$\hat{\underline{\beta}} \sim N(\underline{\beta}; \tau(\hat{\underline{\beta}})) = \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} \sim N\left(\begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}; \tau(\hat{\underline{\beta}})^{-1}\right)$$

$$\hat{\beta}_r \sim N(\beta_r; [\tau(\hat{\underline{\beta}})^{-1}]_{(r;r)})$$

GAUSSIAN $f(y_i; \theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{1}{2\sigma^2} [y_i - \mu]^2\right\}$ $\mu = \beta_1 + \beta_2 x_i$

LIKELIHOOD

$$L(\theta) = \prod \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{1}{2\sigma^2} [y_i - \mu_i]^2\right\} = \prod (2\pi)^{-1/2} \cdot \prod (\sigma^2)^{-1/2} \exp\left\{-\frac{1}{2\sigma^2} \sum (y_i - \mu_i)^2\right\} = (2\pi)^{-n/2} (\sigma^2)^{-n/2} \exp\left\{-\frac{1}{2\sigma^2} \sum (y_i - \mu_i)^2\right\}$$

LOG-LIKELIHOOD

$$l(\theta) = \log(L(\theta)) = \log\left\{ (2\pi)^{-n/2} (\sigma^2)^{-n/2} \exp\left\{-\frac{1}{2\sigma^2} \sum (y_i - \mu_i)^2\right\}\right\} = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum (y_i - \mu_i)^2$$

SCORE FUNCTION

$$I_\theta(\theta) = \begin{cases} a = \frac{d}{d\beta_1} l(\theta) \\ b = \frac{d}{d\beta_2} l(\theta) \\ c = \frac{d}{d\sigma^2} l(\theta) \end{cases}$$

$$a = \frac{d}{d\beta_1} \left[-\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum (y_i - \mu_i)^2 \right] = \frac{d}{d\beta_1} \left[-\frac{1}{2\sigma^2} \sum (y_i - \beta_1 - \beta_2 x_i)^2 \right] = \frac{1}{\sigma^2} \sum 2(y_i - \beta_1 - \beta_2 x_i) = \frac{2}{\sigma^2} \sum (y_i - \beta_1 - \beta_2 x_i)$$

$$b = \frac{d}{d\beta_2} \left[-\frac{1}{2\sigma^2} \sum (y_i - \beta_1 - \beta_2 x_i)^2 \right] = \frac{1}{\sigma^2} \sum 2(y_i - \beta_1 - \beta_2 x_i) x_i = \frac{2}{\sigma^2} \sum (y_i - \beta_1 - \beta_2 x_i) x_i$$

$$c = \frac{d}{d\sigma^2} \left[-\frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum (y_i - \mu_i)^2 \right] = -\frac{n}{2\sigma^2} - \frac{1}{2} \sum (y_i - \mu_i)^2 \cdot \frac{(-1)}{(\sigma^2)^2} = -\frac{n}{2\sigma^2} + \frac{1}{2} \sum (y_i - \mu_i)^2 \frac{1}{\sigma^3}$$

$$\frac{d}{dx} \frac{1}{x} = \frac{d}{dx} x^{-1} = -x^{-2} = -\frac{1}{x^2}$$

$$\hat{\beta} \sim N_p(\beta; \sigma^2 (X^T X)^{-1})$$

$$\hat{\beta}_1 \sim N_p(\beta_1; [\sigma^2 (X^T X)^{-1}]_{(1,1)})$$

Poisson

$$p(y_i; \lambda_i) = \frac{e^{-\lambda_i} \lambda_i^{y_i}}{y_i!} \quad \lambda_i = e^{\tilde{\lambda}_i^T \theta}$$

UNGLIKED

$$L(\theta) = \prod \frac{e^{-\lambda_i} \lambda_i^{y_i}}{y_i!}$$

LOG LIKELIHOOD

$$\begin{aligned} l(\theta) = \log(L(\theta)) &= \log\left(\prod \frac{e^{-\lambda_i} \lambda_i^{y_i}}{y_i!}\right) = \sum \log\left(\frac{e^{-\lambda_i} \lambda_i^{y_i}}{y_i!}\right) = \sum \log(e^{-\lambda_i}) + \log(\lambda_i)^{y_i} - \log(y_i!) \\ &= \sum -\lambda_i + y_i \log(\lambda_i) - \log(y_i!) \\ &= \sum -e^{\tilde{\lambda}_i^T \theta} + y_i \tilde{\lambda}_i^T \theta - \log(y_i!) \end{aligned}$$

SCORE FUNCTION

$$l_{\eta}(\theta) = \begin{cases} a = \frac{d}{d\beta_1} L(\theta) \\ b = \frac{d}{d\beta_2} L(\theta) \end{cases}$$

$$a = \frac{d}{d\beta_1} \left[\sum -e^{\tilde{\lambda}_i^T \theta} + y_i \tilde{\lambda}_i^T \theta - \log(y_i!) \right] = \frac{d}{d\beta_1} \sum -e^{\beta_1 + \beta_2 x_i} + y_i (\beta_1 + \beta_2 x_i) - \log(y_i!) = \sum -e^{\beta_1 + \beta_2 x_i} + y_i$$

$$b = \frac{d}{d\beta_2} \left[\sum -e^{\beta_1 + \beta_2 x_i} + y_i (\beta_1 + \beta_2 x_i) - \log(y_i!) \right] = \sum -e^{\beta_1 + \beta_2 x_i} x_i + y_i x_i$$

$$\bullet \hat{\underline{\beta}} \sim N(\underline{\beta}; \mathcal{I}(\hat{\underline{\beta}})^{-1}) \rightarrow \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} \sim N\left(\begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}; \mathcal{I}(\hat{\underline{\beta}})^{-1}\right)$$

$$\bullet \hat{\beta}_1 \sim N(\beta_1; \mathcal{I}(\hat{\underline{\beta}})^{-1}_{(1;1)})$$

LIKELIHOOD, LOG-LIKELIHOOD, SCORE FUNCTION

$$f(\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(y_i - \mu_i)^2\right\} \quad \mu_i = \beta_1 + \beta_2 x_i$$

LIKELIHOOD

$$L(\theta) = \prod \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(y_i - \mu_i)^2\right\} = \prod \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2} \sum (y_i - \mu_i)^2\right\}$$

LOG-LIKELIHOOD

$$\begin{aligned} l(\theta) = \log(L(\theta)) &= \log\left(\prod \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2} \sum (y_i - \mu_i)^2\right\}\right) \\ &= \log\left(\prod \frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{1}{2\sigma^2} \sum (y_i - \mu_i)^2 = \sum \log\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{1}{2\sigma^2} \sum (y_i - \mu_i)^2 = -\frac{1}{2} \sum \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (y_i - \mu_i)^2 \\ &= -\frac{1}{2} \sum \log(2\pi) - \frac{1}{2} \sum \log(\sigma^2) - \frac{1}{2\sigma^2} \sum (y_i - \mu_i)^2 = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum (y_i - \mu_i)^2 \end{aligned}$$

SCORE FUNCTION

$$l_1(\theta) = \begin{cases} a = \frac{d}{d\beta_1} l(\theta) \\ b = \frac{d}{d\beta_2} l(\theta) \\ c = \frac{d}{d\sigma^2} l(\theta) \end{cases}$$

$$a = \frac{d}{d\beta_1} l(\theta) = \frac{d}{d\beta_1} \left(-\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum (y_i - \mu_i)^2 \right) = -\frac{1}{2\sigma^2} \frac{d}{d\beta_1} \sum (y_i - \beta_1 - \beta_2 x_i)^2 = -\frac{1}{2\sigma^2} \sum 2(y_i - \beta_1 - \beta_2 x_i)(-1)$$

$$b = \frac{d}{d\beta_2} \left(-\frac{1}{2\sigma^2} \sum (y_i - \beta_1 - \beta_2 x_i)^2 \right) = -\frac{1}{2\sigma^2} \sum 2(y_i - \beta_1 - \beta_2 x_i)(-x_i)$$

$$c = \frac{d}{d\sigma^2} l(\theta) = -\frac{n}{2} \frac{1}{\sigma^2} - \frac{1}{2} \sum (y_i - \mu_i)^2 \frac{d}{d\sigma^2} \frac{1}{\sigma^2} = -\frac{n}{2} \frac{1}{\sigma^2} - \frac{1}{2} \left(-\frac{1}{\sigma^2} \right) \sum (y_i - \mu_i)^2$$

$\frac{d}{dx} \frac{1}{x} = \frac{d}{dx} x^{-1} = -x^{-2} = -\frac{1}{x^2}$

$$\hat{\beta}(y) \sim N_p(\hat{\beta}; (X^T X)^{-1} \sigma^2)$$

$$\hat{\beta}(y) \sim N_1(\beta_1; (X^T X)^{-1}_{(1,1)} \sigma^2)$$